DARES: Domain Adaptation via α-Rényi Entropy for Semantic Segmentation

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Abstract

1 Introduction

Forest ecosystems are vital components of the global biosphere, playing a crucial role in climate regulation, carbon sequestration, and biodiversity preservation. Monitoring deforestation and forest degradation via remote sensing imagery is essential for informing environmental policies and conservation strategies. Deep learning techniques, particularly Convolutional Neural Networks (CNNs) like U-Net [34] and its variants [33], have achieved unprecedented accuracy in semantic segmentation tasks for Earth Observation [32]. However, collecting large-scale dense annotations for every new satellite sensor or geographic region is prohibitively expensive [7, 23]. To address this, Unsupervised Domain Adaptation (UDA) has emerged as a promising solution to transfer knowledge from a labeled source domain to an unlabeled target domain [21, 42]. The remote sensing community has rapidly adopted these techniques to tackle various ecological and urban challenges [12, 13]. For instance, adapting segmentation models across different optical sensors heavily relies on aligning multi-scale feature representations [6, 17]. Furthermore, recent studies emphasize the importance of preserving structural information when generalizing across distinct geographic regions [45, 48]. By mitigating domain shifts without requiring target labels, these frameworks enable scalable monitoring solutions [4, 18]. Ultimately, robust UDA strategies are critical for the deployment of deep learning in large-scale Earth Observation pipelines [16]. In recent years, adversarial training

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approaches [19, 38] and entropy minimization techniques [41, 43] have emerged as dominant paradigms for UDA in semantic segmentation. While effective, adversarial methods often struggle with training instability and mode collapse, and standard Shannon entropy minimization can be sensitive to isolated confident misclassifications. Information-theoretic learning, specifically leveraging Rényi's α-entropy [2, 35], provides a robust non-parametric framework for distribution alignment directly from data samples. Inspired by the Conditional Domain Adaptation with α-Rényi Entropy framework [36], we propose DARES (Domain Adaptation via α-Rényi Entropy for Semantic Segmentation), specifically tailored for the highly structured, spatial nature of satellite imagery in forest monitoring tasks. We leverage a Spatially-Stratified, Confidence-Guided Sampling operator to extract semantically rich representations from high-resolution feature maps, optimizing the alignment through a positive definite Gram matrix estimator.

Our main contributions are fourfold:

- We introduce DARES, a novel UDA framework that leverages matrix-based Conditional Rényi α-entropy for semantic segmentation, extending information-theoretic alignment to dense prediction tasks.
- We propose a Spatially-Stratified, Confidence-Guided Sampling operator that ensures diverse and reliable feature extraction while bounding the memory complexity of the Gram matrix computation.
- We adapt the ResUNet backbone to produce robust spatial embeddings suitable for reproducing kernel Hilbert space (RKHS) evaluations.
- We demonstrate the efficacy of our method in an ecologically critical scenario: transferring deforestation mapping models trained on the Brazilian Amazon (PRODES) to the Colombian Forest Reserves.

The remainder of this paper is organized as follows: Section 2 reviews the literature on domain adaptation, semantic segmentation, and information theory. Section 3 details the dataset, the ResUNet backbone, and the proposed DARES methodology. Section 4 presents the experimental setup, followed by results and conclusions in Sections **??** and **??**, respectively.

2 Related Work

The intersection of domain adaptation, remote sensing, and information theory constitutes a rapidly growing field of research [11, 31]. Numerous methods have been proposed to align feature distributions across varying spectral characteristics [29, 30]. For example, several architectures utilize self-training strategies to generate reliable pseudo-labels for unlabeled remote sensing imagery [8, 26]. Others focus on contrastive learning paradigms to enforce consistency between source and target domains [27, 46]. Attention mechanisms and Transformer-based models have also shown exceptional capability in bridging large domain gaps [20,24]. In parallel, multi-task learning frameworks leverage auxiliary tasks such as boundary detection to guide the adaptation process [5, 25]. To counter the effects of complex covariate shifts, researchers have increasingly incorporated frequency-domain filtering techniques [9,28]. Moreover, class-balanced sampling techniques are often integrated to prevent dominant classes from overwhelming the alignment objective [10, 15]. Advanced discrepancy metrics, including Wasserstein distance and maximum mean discrepancy, continue to provide strong theoretical guarantees for UDA [14, 37]. Consequently, the landscape of remote sensing UDA remains diverse and highly active [1]. Table 1 summarizes the prominent unsupervised domain adaptation approaches applied to semantic segmentation.

Adversarial methods, such as ADVENT [38] and CyCADA [19], seek to align distributions by fooling a domain discriminator, either at the feature or pixel level. Self-training methods [43] rely on generating pseudo-labels for the target domain and iteratively retraining the model. Recently, DAFormer [22] demonstrated state-of-the-art performance using Transformer architectures. However, these methods often rely on complex adversarial optimization or heuristic pseudo-label filtering.

Table 1: Comparison of Unsupervised Domain Adaptation Methods for Semantic Segmentation

Method	**Paradigm**	**Key Mechanism**	**Year**
CyCADA [19]	Adversarial	Pixel & Feature-level alignment	2018
CBST [43]	Self-Training	Class-balanced pseudo-labels	2018
ADVENT [38]	Adversarial	Entropy minimization via discriminator	2019
DAFormer [22]	Transformer	Masked Image Modeling	2022
α-Rényi UDA [36]	Info-Theoretic	Matrix-based Rényi α-entropy	2024
DARES (Ours)	Info-Theoretic	Spatially-stratified Rényi alignment	-

In remote sensing, semantic segmentation relies heavily on U-Net variants [34]. The ResUNet architecture [33, 44], which integrates residual connections into the U-Net encoder-decoder structure, has proven particularly effective for Earth Observation tasks, preserving both high-level semantic context and fine-grained spatial details.

Our work builds upon the theoretical foundation of Information Theoretic Learning (ITL) [35]. Unlike Shannon entropy, which requires explicit density estimation, the matrix-based formulation of Rényi's α-entropy [3] allows for direct estimation of information measures from the eigenspectrum of Gram matrices in an RKHS. This enables robust and differentiable optimization of divergence and conditional entropy [2, 36], which we extend to dense spatial tasks in this work.

Table 2: Evolution of Semantic Segmentation Architectures in Earth Observation

Architecture	Core Innovation	Application Focus
U-Net [34]	Skip connections	Medical / General
SegNet [39]	Max-pooling indices	Autonomous Driving
TernausNet [40]	Pre-trained VGG encoder	High-Res Satellite
ResUNet [44]	Residual blocks in U-Net	Road/Forest mapping
ResUNet-a [33]	Dilated convolutions, multi-task	Very High-Res Imagery

3 Materials and Methods

3.1 Datasets

To validate our domain adaptation framework in an ecologically critical context, we utilize two distinct satellite imagery datasets focusing on deforestation monitoring. The **Source Domain** comprises data from the Brazilian Amazon (PRODES dataset), a large-scale, well-annotated repository tracking deforestation patterns. The **Target Domain** consists of high-resolution imagery from the Colombian Forest Reserves, which presents a significant domain shift due to variations in sensor characteristics, atmospheric conditions, and distinct regional topography. The task is a binary semantic segmentation mapping: Forest vs. Deforestation/Non-Forest.

3.2 Deep Learning Backbone (ResUNet)

We employ the ResUNet architecture as our deep learning backbone. ResUNet effectively combines the encoder-decoder structure of U-Net with residual connections, facilitating the learning of both high-level semantic features and precise spatial localization without suffering from vanishing gradients. The network acts as a feature extractor $F : \mathcal{X} \rightarrow \mathbb{R}^{D\times H\times W}$ mapping an input image x to a dense spatial feature map, followed by a pixel-wise classification head $G : \mathbb{R}^{D\times H\times W} \rightarrow [0, 1]^{C\times H\times W}$, where $C = 2$ corresponds to our class set.

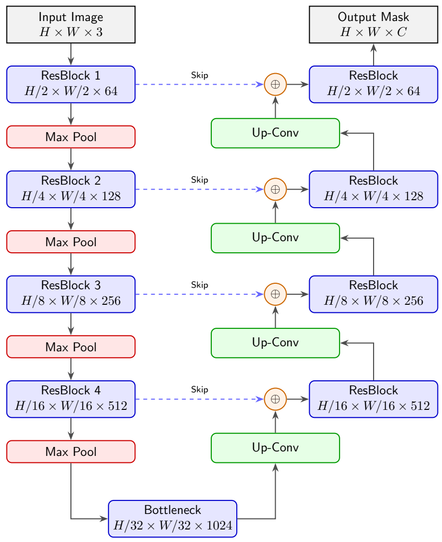

Figure 1: Architecture of the Shared ResUNet Backbone, detailing the encoder-decoder structure with residual blocks and skip connections.

3.3 DARES Methodology

Our objective is to align the class-conditional distributions of the source and target domains in the latent space $\mathbb{R}^D$ using Information Theoretic Learning. Unlike standard image classification, semantic segmentation produces dense feature maps containing millions of pixels per image, making the computation of full Gram matrices $\mathcal{O}((BHW)^2)$ computationally intractable.

To resolve this, we introduce a **Spatially-Stratified, Confidence-Guided Sampling Operator** Φ_c. For a given class $c \in \{1, \ldots, C\}$, Φ_c grid-stratifies the spatial dimensions of the target predictions and selects features whose pseudo-label confidence exceeds a threshold $\tau > 0.85$, capped at $N_{max} = 1024$ samples per class per mini-batch.

Let the sampled source features be $\mathbf{F}_c^s \in \mathbb{R}^{N_c^s \times D}$ and the target features be $\mathbf{F}_c^t \in \mathbb{R}^{N_c^t \times D}$. We construct the class-wise intra-domain and cross-domain Gram matrices using a Gaussian kernel $k(x, y) = \exp(-\|x - y\|^2 / 2\sigma^2)$:

$$\mathbf{K}_c^s = [k(f_i^s, f_j^s)] \in \mathbb{R}^{N_c^s \times N_c^s}, \tag{1}$$

$$\mathbf{K}_c^t = [k(f_i^t, f_j^t)] \in \mathbb{R}^{N_c^t \times N_c^t}, \tag{2}$$

$$\mathbf{K}_c^{st} = [k(f_i^s, f_j^t)] \in \mathbb{R}^{N_c^s \times N_c^t}. \tag{3}$$

To mitigate the impact of noisy target pseudo-labels, we apply a confidence weighting matrix $\tilde{\mathbf{W}}_c^t = w_c(w_c)^\top$, where w_c contains confidence scores (computed as one minus the normalized uncertainty score derived from Rényi's quadratic entropy of the pixel-wise predictions). The re-weighted target matrix is $\tilde{\mathbf{K}}_c^t = \mathbf{K}_c^t \odot \tilde{\mathbf{W}}_c^t$.

The matrix-based estimator for the α-Rényi ($\alpha = 2$) mutual information between the source and target distributions for class c is:

$$\tilde{I}_2(\mathbf{K}_c^s; \tilde{\mathbf{K}}_c^t) = \frac{1}{2}\Big(H_2(\mathbf{K}_c^s) + H_2(\tilde{\mathbf{K}}_c^t) - H_2(\mathbf{K}_c^{mix})\Big), \tag{4}$$

where $\mathbf{K}_c^{mix} = \begin{pmatrix} \mathbf{K}_c^s & \mathbf{K}_c^{st} \\ (\mathbf{K}_c^{st})^\top & \tilde{\mathbf{K}}_c^t \end{pmatrix}$ and $H_2(\mathbf{A}) = -\log(\mathrm{tr}(\tilde{\mathbf{A}}^\top \tilde{\mathbf{A}}))$ with $\tilde{\mathbf{A}} = \mathbf{A}/\mathrm{tr}(\mathbf{A})$.

The total DARES objective combines the supervised segmentation loss (cross-entropy) on the source domain with the class-conditional Rényi alignment:

$$\mathcal{L}_{DARES} = \mathcal{L}_{CE}(\mathcal{D}_s) - \lambda \sum_{c=1}^{C} \tilde{I}_2(\mathbf{K}_c^s; \tilde{\mathbf{K}}_c^t). \tag{5}$$

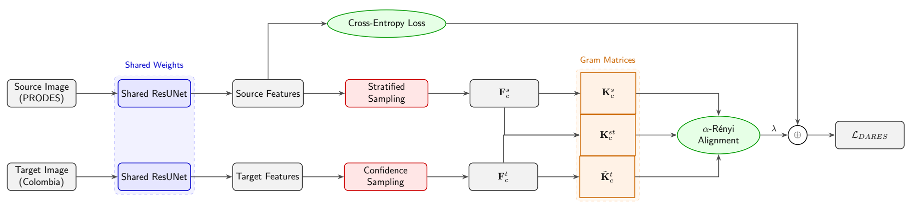

Figure 2: The DARES alignment pipeline. The shared ResUNet processes source and target inputs. Stratified and confidence-guided sampling extract representative feature vectors ($\mathbf{F}_c^s$, $\mathbf{F}_c^t$). These construct the Gram matrices which are optimized via the α-Rényi Alignment Loss.

4 Experimental Setup

The ResUNet backbone is pretrained on the PRODES source dataset. During the unsupervised domain adaptation phase, we jointly train the model on labeled source patches and unlabeled target patches from the Colombian Forest Reserves. We use a batch size of $B = 8$, random crops of 512×512 pixels, and optimize the network using the Adam optimizer with a learning rate of 10^{-4}. The hyperparameter λ balancing the alignment loss is empirically set to 0.1.

To evaluate the performance, we employ the mean Intersection over Union (mIoU) and the F1-score. Let $\mathbf{Y}, \hat{\mathbf{Y}} \in \{0,1\}^{C \times HW}$ represent the one-hot encoded ground truth and predicted segmentation tensors, respectively, flattened across the spatial dimensions. We compute the confusion matrix $\mathbf{M} \in \mathbb{N}^{C \times C}$ through the tensor inner product $\mathbf{M} = \mathbf{Y}\hat{\mathbf{Y}}^\top$.

Using matrix operations, we express the metrics for all classes simultaneously as vectors $\mathbf{iou}, \mathbf{f1} \in \mathbb{R}^C$. Let $\mathbf{1} \in \mathbb{R}^C$ be a column vector of ones, and $\mathrm{diag}(\mathbf{M})$

denote the column vector formed by the main diagonal of $\mathbf{M}$. The mathematical expressions in tensor matrix form are given by:

$$\mathbf{iou} = \mathrm{diag}(\mathbf{M}) \oslash \left(\mathbf{M1} + \mathbf{M}^{\top}\mathbf{1} - \mathrm{diag}(\mathbf{M})\right), \tag{6}$$

$$\mathbf{f1} = 2 \odot \mathrm{diag}(\mathbf{M}) \oslash \left(\mathbf{M1} + \mathbf{M}^{\top}\mathbf{1}\right), \tag{7}$$

where $\oslash$ and $\odot$ represent the Hadamard (element-wise) division and multiplication, respectively. The overall mIoU is calculated as $\frac{1}{C}\mathbf{1}^{\top}\mathbf{iou}$. We compare DARES against standard baselines, including a 'Source-Only' model (no adaptation) and adversarial alignment methods such as ADVENT [38].

5 Results and Discussion

6 Conclusions

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